

# DHCP Snooping

|                                                         |   |
|---------------------------------------------------------|---|
| <a href="#">Server valide</a>                           | 0 |
| <a href="#">configurer le switch avec snooping</a>      | 3 |
| <a href="#">Test avec le serveur DHCP valide éteint</a> | 4 |
| <a href="#">Vérification sur le switch</a>              | 4 |

## Server valide

Server DHCP valide

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameDHCP\_valide

Default Gateway192.168.1.254

DNS Server0.0.0.0

Start IP Address : 1921681100

Subnet Mask: 2552552550

Maximum Number of Users : 10

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

AddSaveRemove

| Pool Name   | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|-------------|-----------------|------------|------------------|-------------|----------|-------------|-------------|
| serverPool  | 0.0.0.0         | 0.0.0.0    | 192.168....      | 255.255.... | 0        | 0.0.0.0     | 0.0.0.0     |
| DHCP_valide | 192.168....     | 0.0.0.0    | 192.168....      | 255.255.... | 10       | 0.0.0.0     | 0.0.0.0     |

Top

PC 1

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.101

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

## PC2

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.102

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

IPv6 Configuration

## Server invalide

Server DHCP de la personne malveillante

Physical
Config
**Services**
Desktop
Programming
Attributes

**SERVICES**

HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP

DHCP

Interface
FastEthernet0
Service
☒ On
☐ Off

Pool Name
DHCP\_invalide

Default Gateway
172.17.0.254

DNS Server
0.0.0.0

Start IP Address :
172
17
30
100

Subnet Mask:
255
255
0
0

Maximum Number of Users :
10

TFTP Server:
0.0.0.0

WLC Address:
0.0.0.0

Add
Save
Remove

| Pool Name     | Default Gateway | DNS Server | Start IP Address | Subnet Mask | Max User | TFTP Server | WLC Address |
|---------------|-----------------|------------|------------------|-------------|----------|-------------|-------------|
| serverPool    | 0.0.0.0         | 0.0.0.0    | 172.17.0.0       | 255.255.... | 0        | 0.0.0.0     | 0.0.0.0     |
| DHCP_invalide | 172.17.0...     | 0.0.0.0    | 172.17.3...      | 255.255.... | 10       | 0.0.0.0     | 0.0.0.0     |

☐ Top

## PC 1

IP Configuration

☒ DHCP
☐ Static
DHCP request successful.

IPv4 Address
172.17.30.101

Subnet Mask
255.255.0.0

Default Gateway
172.17.0.254

DNS Server
0.0.0.0

## PC 2

| IP Configuration                      |                              |
|---------------------------------------|------------------------------|
| <input checked="" type="radio"/> DHCP | <input type="radio"/> Static |
| DHCP request successful.              |                              |
| IPv4 Address                          | 172.17.30.100                |
| Subnet Mask                           | 255.255.0.0                  |
| Default Gateway                       | 172.17.0.254                 |
| DNS Server                            | 0.0.0.0                      |

## configurer le switch avec snooping

Activer le DHCP Snooping globalement sur le switch

```
ip dhcp snooping
```

Spécifier pour quel VLAN le snooping doit être actif

```
ip dhcp snooping vlan 1
```

Aller sur l'interface où est connecté le SERVEUR DHCP VALIDE

```
interface FastEthernet0/24
```

Déclarer ce port comme un port de confiance. Seuls les ports de confiance peuvent envoyer des messages DHCP

```
ip dhcp snooping trust
```

Limiter le nombre de paquets DHCP par seconde pour éviter les attaques par déni de service

```
ip dhcp snooping limit rate 100
```

## Test avec les 2 dhcp

Avec les deux dhcp activée seul le dhcp valide est utilisé

## PC1

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

## PC2

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.101

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

## Test avec le serveur DHCP valide éteint

le service DHCP n'est plus disponible

## PC1

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.156.145

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

## PC2

IP Configuration

☒ DHCP ☐ Static DHCP failed. APIPA is being used.

IPv4 Address 169.254.121.196

Subnet Mask 255.255.0.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

## Vérification sur le switch

```
show ip dhcp snooping
```

on retrouve bien l'interface ou le snooping est actif et la limite de temps

```
Switch#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
1
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                Trusted      Rate limit (pps)
-----
FastEthernet0/3          no          unlimited
FastEthernet0/1          no          unlimited
FastEthernet0/2          no          unlimited
FastEthernet0/24         yes         100
Switch#
```

```
show ip dhcp snooping binding
```

on retrouve les 2 ip distribué

```
Switch#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
00:0D:BD:05:79:C4  192.168.1.101  86400      dhcp-snooping  1     FastEthernet0/1
00:09:7C:51:9C:91  192.168.1.102  86400      dhcp-snooping  1     FastEthernet0/2
Total number of bindings: 2
Switch#
```